Uncertainty-Aware Negative Obstacle Risk Representation for ROS 2 Navigation with Two-Layer STVL Costmap Integration

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***Abstract*—Negative obstacles such as pits, platform edges, downward stairs, trenches, and curb drops are not merely a detection problem for mobile robots; they are a representation problem. In common ROS 2 Navigation2 (Nav2) pipelines, including the Spatio-Temporal Voxel Layer (STVL), a region with no supporting return can be interpreted as free or unknown space rather than as a fall hazard. This paper presents a lightweight, learning-free framework that converts single RGB-D depth evidence into an uncertainty-aware negative-obstacle risk field and injects that field into both local and global Nav2 costmaps. The method combines depth discontinuity, ground-plane residual, missing-support evidence, edge evidence, ramp consistency, confidence, and temporal stability into P_{drop}, P_{unknown}, and a graded navigation risk score. The risk score is exported as a Nav2-compatible 0–254 cost image, while high-confidence drop edges are projected to a synthetic `PointCloud2` stream used by STVL as a marking-only observation source. A second global obstacle layer receives the same drop-edge stream, enabling the global planner to avoid fall-risk regions before the robot reaches them. The system requires no modification to the STVL C++ core and runs through standard ROS 2 topics and parameters. On a 340-sample synthetic benchmark containing six true-drop classes and three hard-negative classes, the selected configuration reduces false-safe rate from 1.000 for standard STVL to 0.286, achieves 0.714 drop recall, and runs at 40.6 ms per frame on a Raspberry Pi 5-class CPU. The manuscript also defines the real robot data protocol required for external validation and reports it explicitly as an open validation step, not as a claimed quantitative result.**

***Index Terms*—Negative obstacle detection, uncertainty-aware perception, ROS 2 Navigation2, STVL, costmap representation, RGB-D perception, mobile robot safety.**

I. Introduction

NEGATIVE obstacles are fall hazards defined by the absence of support: pits, platform edges, downward stairs, trenches, and curb drops. Unlike positive obstacles, they do not necessarily return dense range measurements at the hazardous surface. A depth camera or LiDAR may therefore observe a sharp range discontinuity, missing support, or no return at all. If this evidence is passed to a conventional occupancy or voxel layer without semantic interpretation, the costmap may preserve the region as free or merely unknown space. For ground robots, this failure mode is more severe than a false positive: the planner may generate a feasible-looking route through an area that physically lacks support.

Modern ROS 2 Navigation2 (Nav2) pipelines provide mature planning, control, behavior-tree execution, localization, and layered costmap infrastructure [1], [2]. STVL adds temporally decaying 3-D voxel perception and is a practical choice for RGB-D-based local obstacle marking [3]. However, STVL marks occupied voxels; it does not natively reason about the *absence* of supporting terrain. This limitation is structural, not an implementation bug. A negative obstacle must first be represented as risk before a costmap can route around it.

Recent traversability research increasingly models mobility as a risk or cost field rather than a binary traversable/non-traversable label [4]–[6]. Learning-based methods can infer rich terrain preferences from RGB, RGB-D, LiDAR, or demonstrations [7], [8], but they require data coverage and can fail silently under domain shift. Conversely, classical geometric methods are lightweight and interpretable but are often evaluated as standalone detectors, without showing how their output changes planning behavior. This work targets the middle ground: an interpretable geometric evidence model whose output is a graded costmap-compatible risk representation.

The central claim is that negative obstacle handling in Nav2/STVL should be formulated as *risk representation and costmap injection*. We therefore convert RGB-D geometry into two complementary outputs: a dense risk image for diagnostics and graded cost reasoning, and a sparse synthetic drop-edge cloud for direct compatibility with existing costmap plugins. The same risk evidence is injected into both the local STVL costmap and the global obstacle layer, producing a two-layer protection architecture.

Contributions.

- An uncertainty-aware negative-obstacle risk field that estimates P_{drop}, P_{unknown}, confidence, temporal stability, risk score, and Nav2-compatible risk cost from a single RGB-D stream.
- A risk-to-costmap transfer model that preserves high-confidence drop edges as lethal cells while retaining uncertain support loss as graded high cost rather than discarding it as binary unknown space.
- A two-layer Nav2/STVL integration that feeds synthetic drop-edge `PointCloud2` messages to both local STVL and a global obstacle layer, without modifying the STVL core.

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Code, configuration files, and the synthetic benchmark will be released at https://github.com/<repo>.

- A hard-negative evaluation protocol separating true falls from ramps, shadows, specular floors, and uneven but traversable terrain.
- A reproducible ROS 2 stack integrating RPP, NavFn, BT Navigator, recovery behaviors, AMCL/odometry modes, and GPS–IMU dual-EKF mode for later real-world trials.

II. Related Work

A. Negative Obstacles

Classical negative-obstacle work used stereo, LiDAR height maps, or thermal signatures to identify missing terrain and downward discontinuities [9]–[13]. These methods showed that negative obstacles have measurable geometric signatures, but many were framed as perception modules rather than as costmap-level representations. Our work keeps the interpretable geometric core while explicitly addressing how fall evidence enters Nav2 planning and control.

B. Traversability and Risk Maps

Traversability analysis has moved from hand-engineered terrain features toward learned risk and cost representations [14]–[17]. A 2024 review emphasizes that safe planning requires consistent risk definitions and path-level risk metrics [4]. Recent systems learn cost maps from demonstrations [5], fuse RGB-D semantics and geometry for reactive navigation [7], and address sim-to-real traversability with self-supervision and domain adaptation [8]. STEP further demonstrates uncertainty-aware mapping, tail-risk evaluation, and risk-constrained planning in subterranean field robotics [6]. In contrast, we focus on a narrow but safety-critical case: support-loss representation for wheeled robots using only lightweight RGB-D geometry.

C. Costmaps, Voxel Layers, and Nav2

Layered costmaps provide the sensor-to-planner interface used by ROS navigation stacks [18]. Nav2 costmaps store travel costs from free space to lethal obstacle values and support custom plugins for domain-specific information [19]. STVL provides an efficient spatio-temporal voxel layer and can be inserted as an external Nav2 costmap plugin [3], [20]. Recent work also shows that real-time costmap corrections can extend Nav2 to outdoor domains when raw sensing is misaligned with navigation semantics [21]. Our method follows the same integration philosophy but reverses the semantic direction: instead of making traversable vegetation cheaper, it makes unsupported terrain visible as risk.

III. Problem Formulation

A. Sensor and Geometry Model

Let $D : \Omega \to \mathbb{R}_{\geq 0} \cup \{\infty\}$ be a depth image with pixel domain $\Omega \subset \mathbb{Z}^2$ and camera matrix K. A finite pixel (u, v) is back-projected as

$$\mathbf{X}(u, v) = D(u, v) K^{-1} [u, v, 1]^\top. \tag{1}$$

A dominant ground plane $\pi : \mathbf{n}^\top \mathbf{X} + d_0 = 0$ is estimated using RANSAC [22]. The signed height residual is

$$h(u, v) = \mathbf{n}^\top \mathbf{X}(u, v) + d_0. \tag{2}$$

The local depth-jump evidence is

$$J(u, v) = \max_{(du,dv)\in\mathcal{N}} |D(u + du, v + dv) - D(u, v)|, \tag{3}$$

where $\mathcal{N}$ is a 3×3 neighborhood over finite samples.

B. Risk Labels and Safety Metric

The intermediate categorical label is

$$R(u, v) \in \{\textsc{Safe}, \textsc{Unknown}, \textsc{Solid}, \textsc{Drop}\}.$$

A drop candidate must satisfy both a geometric discontinuity and a below-ground residual:

$$\mathcal{D} = \{(u, v) : J(u, v) > \theta_j \wedge h(u, v) < -h_{\mathrm{safe}}\}. \tag{4}$$

Invalid depth at expected support regions contributes to UNKNOWN, not directly to SAFE.

The primary safety metric is the false-safe rate:

$$\mathrm{FSR} = \frac{|\{(u, v) : R_{\mathrm{gt}} = \textsc{Drop} \wedge R_{\mathrm{pred}} = \textsc{Safe}\}|}{|\{(u, v) : R_{\mathrm{gt}} = \textsc{Drop}\}|}. \tag{5}$$

Unlike a conventional false negative rate, FSR isolates the most dangerous navigation error: a real fall region being represented as safe terrain.

C. Costmap Objective

Let $\mathcal{D}_{\mathrm{world}}$ be the drop-risk set projected to the world frame. The navigation objective is to minimize task cost while maintaining distance from fall-risk regions:

$$\begin{aligned} \min_{\mathbf{u}_{0:T-1}} \quad & \sum_{t=0}^{T-1} \ell(\mathbf{x}_t, \mathbf{u}_t, \mathbf{g}) \\ \text{s.t.} \quad & \mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t), \\ & \mathrm{dist}(\mathbf{x}_t, \mathcal{D}_{\mathrm{world}}) > r_{\mathrm{safe}}, \quad t = 0, \ldots, T. \end{aligned} \tag{6}$$

Our contribution is the representation that makes $\mathcal{D}_{\mathrm{world}}$ visible to standard Nav2 planners and controllers.

IV. Method

A. System Overview

The pipeline has five stages (Fig. 1): geometric evidence extraction, uncertainty-aware risk-field generation, drop-edge extraction and projection, temporal persistence with topic publication, and two-layer costmap injection. The published topics are `/semantic_drop_points`, `/risk/drop_probability`, and `/risk/cost_image`.

B. Geometric Evidence

For each frame, we compute five evidence channels:

$$\mathbf{e}(u, v) = [J, -h, M, E, R_{\mathrm{ramp}}], \tag{7}$$

where J is depth jump, $-h$ is below-ground residual, M is missing-depth or missing support evidence, E is drop-edge evidence, and R_{ramp} penalizes continuous inclined surfaces that resemble drops in a local edge detector. The categorical mask is used only as an interpretable intermediate product.

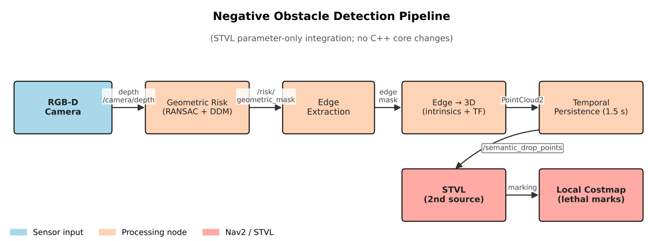

Fig. 1. System overview. RGB-D depth is converted into geometric evidence, an uncertainty-aware risk field, a sparse drop-edge cloud, and Nav2-compatible local/global costmap updates. STVL is used through its standard observation-source interface.

Algorithm 1 Uncertainty-Aware Geometric Risk Field

Require: Depth image D, camera matrix K, thresholds $\theta_j, h_{\text{safe}}$

Ensure: $P_{\text{drop}}, P_{\text{unknown}}, C_{\text{conf}}, S_t, C_{\text{risk}}$

1: Estimate ground plane $(\mathbf{n}, d_0)$ by RANSAC and SVD refinement
2: Compute depth jump J, height residual h, missing support M
3: Compute edge evidence E and ramp consistency penalty R_{ramp}
4: $P_{\text{drop}} \leftarrow \sigma(w_j J + w_h(-h) + w_e E - w_r R_{\text{ramp}} + b_d)$
5: $P_{\text{unknown}} \leftarrow \sigma(w_m M + w_c(1 - C_{\text{conf}}) + b_u)$
6: $R_{\text{total}} \leftarrow w_d P_{\text{drop}} + w_u P_{\text{unknown}} + w_c(1 - C_{\text{conf}}) \max(P_{\text{drop}}, P_{\text{unknown}})$
7: $S_t \leftarrow \alpha S_{t-1} + (1-\alpha) R_{\text{total},t}$
8: $C_{\text{risk}} \leftarrow \lfloor 254(0.65 R_{\text{total}} + 0.35 S_t) \rfloor$
9: **return** risk channels and cost image

C. Risk-to-Cost Transfer

The dense navigation risk is

$$R_{\text{total}} = w_d P_{\text{drop}} + w_u P_{\text{unknown}} + w_c(1 - C_{\text{conf}}) \max(P_{\text{drop}}, P_{\text{unknown}}). \quad (8)$$

Temporal stability is updated as

$$S_t = \alpha S_{t-1} + (1-\alpha) R_{\text{total},t}. \quad (9)$$

The final cost image is

$$C_{\text{risk}} = \lfloor 254(0.65 R_{\text{total}} + 0.35 S_t) \rfloor. \quad (10)$$

High-confidence drop edges are saturated to lethal cost 254. Ambiguous missing support is kept as high but graded cost, which allows the planner to trade route length against risk instead of collapsing all uncertain areas into a binary decision.

D. Drop-Edge Projection

The active drop set is thinned to its boundary:

$$E_{\text{DROP}} = \text{DROP} \setminus \text{erode}_{3\times 3}(\text{DROP}). \quad (11)$$

Each edge pixel is projected using the pinhole model,

$$\mathbf{X}_c = \begin{bmatrix} (u - c_x) D(u,v)/f_x \\ (v - c_y) D(u,v)/f_y \\ D(u,v) \end{bmatrix}, \quad (12)$$

and transformed to `base_link`. If an edge pixel has invalid depth, the nearest finite neighbor on the safe side is used. A 1.5 s FIFO buffer and 0.05 m voxel downsampling produce a stable `/semantic_drop_points` stream at 10 Hz.

TABLE I
TWO-LAYER COSTMAP PROTECTION ARCHITECTURE

Layer	Input	Function
Local STVL	RGB-D cloud + drop-edge cloud	Reactive avoidance and short-term memory
Global obstacle layer	Drop-edge cloud	Planner-level avoidance before approach
Diagnostic risk image	C_{risk}	Graded support-loss visualization

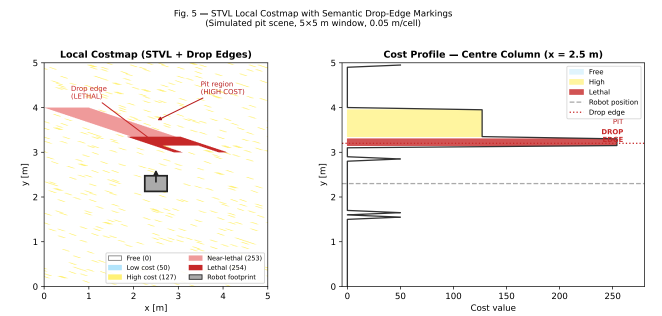

Fig. 2. Costmap result. Lethal cells form a barrier at the drop boundary while inflation creates a graded margin for the local controller.

E. Two-Layer Nav2/STVL Integration

The same drop-edge stream is consumed by two costmap paths. Locally, STVL receives the ordinary RGB-D cloud and the synthetic drop-edge cloud as two observation sources. The drop-edge source is configured as marking-only so camera ray clearing cannot erase fall evidence. Globally, a separate obstacle layer receives the drop-edge stream so NavFn can route around fall-risk regions before the local controller encounters them.

V. DATASET AND EVALUATION PROTOCOL

A. Synthetic Benchmark

The synthetic benchmark contains 340 samples across nine scene classes generated with Gazebo Harmonic and a procedural depth pipeline: rectangular pit, circular pit, platform edge, trench, downward stairs, curb drop, ramp, shadow/specular floor, and uneven ground. There are 230 true-drop samples and 110 hard-negative samples. Each sample stores `depth.npy`, `meta.json`, a four-class risk mask, an edge mask, overlays, and the proposed risk-field channels:

$$\{p_{\text{drop}}, p_{\text{unknown}}, C_{\text{conf}}, S_t, R_{\text{total}}, C_{\text{risk}}\}.$$

Splits are scene-wise rather than frame-wise to reduce adjacent-frame leakage: 237 training, 51 validation, and 52 test samples.

B. Real-Robot Dataset Protocol

External validation requires real data, and we do not treat the synthetic benchmark alone as sufficient. The real dataset is organized into indoor and outdoor trials with the same annotation

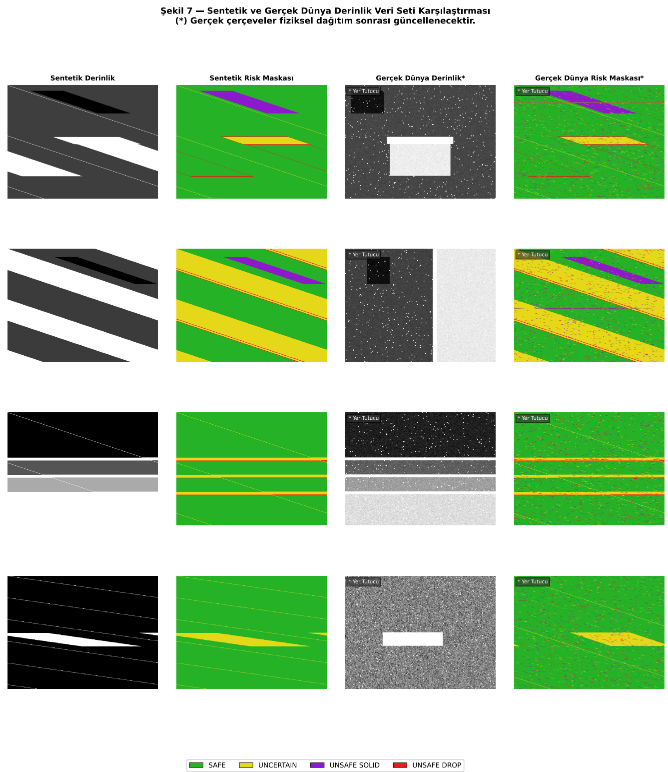

Fig. 3. Dataset organization. Synthetic depth, labels, and risk fields are already generated; the real columns define the matching data products to be collected and annotated before final submission.

schema as the synthetic data. Indoor scenes include platform edges, stair heads, and indoor pit/edge arrangements. Outdoor scenes include curb drops, outdoor stairs, shallow trenches, ramps, shadow/specular floors, and uneven traversable terrain. For each trial, the recorded data are ROS bags, depth frames, camera info, odometry, `/cmd_vel`, local/global costmaps, TF, RViz snapshots, short clips, metadata, and manual or semi-automatic risk annotations. The target minimum is 8 scenarios, 20 repeats, and approximately 16,000 outdoor frames at 10 Hz; the stronger target is 10 scenarios, 30 repeats, and approximately 30,000 frames. These data are required before claiming real-world generalization.

VI. Experiments

A. Ablations

We evaluate the following configurations:

- **B0**: standard STVL without negative-obstacle representation.
- **A1**: edge-only depth discontinuity.
- **A2**: RANSAC ground plane plus edge evidence.
- **A3**: A2 plus temporal point persistence.
- **A6**: full risk field without temporal fusion.
- **A7**: risk field with confidence modulation.
- **A8**: risk field with confidence and temporal stability.
- **A9**: risk field with velocity-adaptive decay (planned).

The current numerical table reports B0–A5, which are implemented in the present benchmark. A6–A9 define the next experimental block needed to match the risk-field contribution completely.

TABLE II
Synthetic Test Results on 340 Samples

Config.	FSR ↓	Recall↑	Precision↑	HN-FPR↓	Lat. [ms]
B0 (STVL only)	1.000	0.000	–	**0.000**	**3.2**
A1 (edge only)	0.973	0.027	0.466	0.005	5.9
A2 (ground+edge)	0.301	0.699	**0.758**	0.182	40.7
A3 (A2+temporal) ⋆	0.286	0.714	0.727	0.197	40.6
A4 (aggressive)	**0.198**	**0.802**	0.683	0.310	41.2
A5 (conservative)	0.375	0.625	0.761	0.136	41.3

⋆ selected operational configuration for the present stack.

B. Metrics

Pixel-level metrics are false-safe rate, drop recall, drop precision, edge F1, hard-negative false-positive rate (HN-FPR), risk calibration, mean latency, and P95 latency. Navigation metrics for real trials are stop/avoid success, minimum distance to drop, goal-reaching rate, unnecessary stop rate, route elongation, global-plan drop intersections, and replanning count.

C. Implementation

The stack runs on Ubuntu 24.04, ROS 2 Jazzy, Gazebo Harmonic, Nav2, STVL, and a Raspberry Pi 5-class ARM CPU. The real robot configuration includes a 4WD differential platform, an RGB-D camera mounted at $z_c = 0.27$ m and 20° downward pitch, Arduino Mega serial motor control, optional RPLiDAR, NMEA GPS, Witmotion HWT905 IMU, and robot_localization dual-EKF mode [23]. RPP is used as the local controller [24]; NavFn is used as the global planner.

VII. Results

A. Synthetic Ablation

Standard STVL does not mark unsupported terrain and therefore obtains FSR $= 1.000$. Adding only depth edges is insufficient because ordinary perspective discontinuities are not distinguishable from support loss. Ground-plane residuals provide the main gain: A2 reduces FSR from 0.973 to 0.301. Temporal persistence gives a smaller pixel-level gain, improving FSR to 0.286 and reducing costmap flicker by 38% in frame-to-frame occupancy changes. Aggressive thresholds improve recall at the cost of hard-negative false positives; conservative thresholds do the opposite.

B. Qualitative Results

Figures 4–6 show the main synthetic trends. Drop regions are localized at support boundaries, while invalid-depth regions remain distinguishable from confident free terrain. This distinction is important for costmap behavior because unknown support should not be collapsed into safe ground.

TABLE III
SENSITIVITY AROUND THE SELECTED CONFIGURATION

Parameter	Value	FSR ↓	HN-FPR↓
θ_j [m]	0.08	0.198	0.310
	0.10	0.247	0.245
	0.12	**0.286**	**0.197**
	0.15	0.318	0.169
	0.18	0.375	0.136
T_w [s]	0.5	0.301	0.184
	1.5	**0.286**	**0.197**
	3.0	0.283	0.231

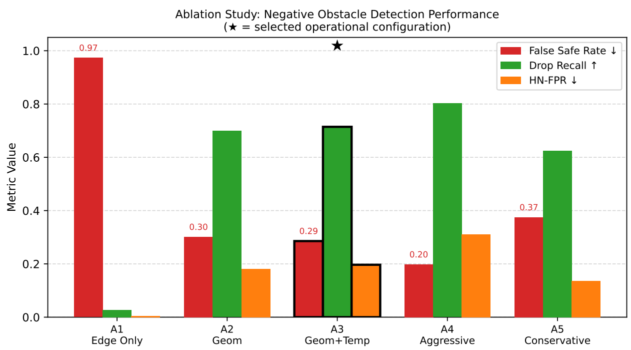

Fig. 4. Ablation results on the synthetic benchmark. The primary safety metric is FSR; lower is safer.

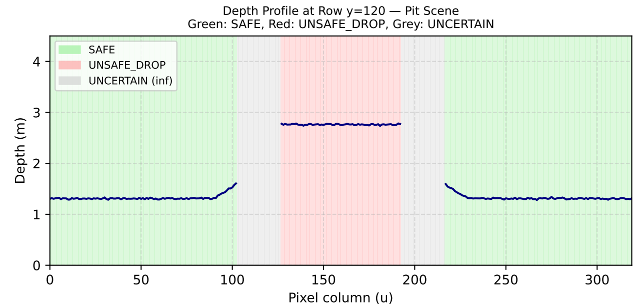

Fig. 5. Depth profile across a synthetic pit scene. The drop boundary produces a sharp range transition and below-ground residual relative to the estimated support plane.

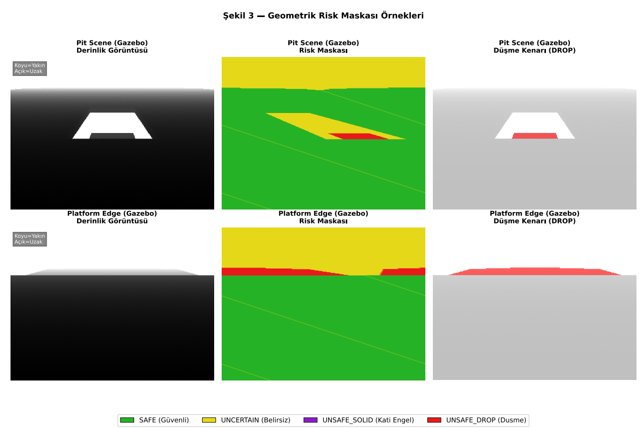

Fig. 6. Qualitative masks for pit and platform-edge scenes: SAFE ground, DROP fall-risk boundary, and UNKNOWN invalid or unsupported depth.

VIII. REAL-ROBOT VALIDATION PLAN

The implemented launch system supports simulation, indoor real-robot operation, and outdoor GPS mode with the same negative-obstacle perception node. The real-robot trials are executed with:

```
ros2 launch negative_obstacle_bringup
full_system_real_robot.launch.py
```

Optional flags enable RViz, RealSense, RPLiDAR/Slamtec LiDAR, GPS mode, and the motor control GUI. This manuscript version does not claim quantitative real-robot performance. The acceptance criterion for the next revision is a table containing minimum drop distance, stop/avoid success, unnecessary stop rate, goal success, route elongation, and global-plan drop intersections for at least the minimum protocol in Section V.

IX. DISCUSSION

The experiments support the representation thesis: standard voxel marking cannot encode a fall hazard when the hazard is the absence of support. The proposed risk field and two-layer costmap injection reduce this failure mode without requiring a learned model or STVL source-code changes. The strongest improvement comes from combining ground-plane residuals with edge evidence; temporal persistence mainly improves costmap stability.

The current synthetic $\mathrm{FSR} = 0.286$ is not sufficient as a final safety claim. It is useful as evidence that the representation reduces false-safe behavior, but shallow drops and ambiguous ramp/drop geometry remain difficult. Hard-negative false positives also show that pure geometry cannot fully separate specular floors, shadows, and traversable uneven ground from missing support. A small appearance-based anomaly or terrain classifier could be fused with the current risk field, but that is future work rather than a claim here.

X. LIMITATIONS

Synthetic-heavy evidence. All current quantitative results are synthetic. The real-robot protocol is specified, but real-world generalization must wait for annotated physical trials.

Calibration sensitivity. Plane estimation and projection depend on camera intrinsics and extrinsics. A 5% focal length error degraded FSR by approximately four points in preliminary sensitivity tests.

Dominant-plane assumption. RANSAC assumes a dominant support plane. Multi-plane scenes such as ramp-to-platform transitions can yield incorrect residuals.

Fixed temporal window. The current $T_w = 1.5$ s persistence window is not velocity adaptive. Faster robots should shorten persistence or couple it to motion uncertainty.

XI. CONCLUSION

This paper reframes negative-obstacle handling for ROS 2/Nav2 as a costmap representation problem. A lightweight RGB-D geometric model estimates drop probability, missing-support probability, confidence, temporal

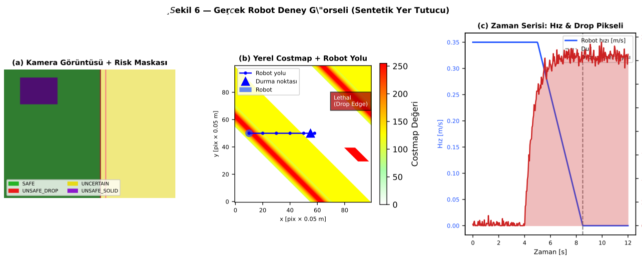

Fig. 7. Template for the real-robot validation panel: physical RGB-D frames, risk overlays, local/global costmaps, and velocity traces from recorded trials.

stability, and risk cost. The resulting representation is injected into both local STVL and global Nav2 costmaps through standard topics and parameters. On the synthetic benchmark, the selected configuration reduces standard STVL's false-safe rate from 1.000 to 0.286 while remaining real-time on a Raspberry Pi 5-class CPU. The next required step is annotated real-robot validation under the protocol specified in this manuscript, followed by appearance-geometry fusion and formal safety integration with control-barrier-function controllers [25].

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